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ORIGINAL PAPER

Factors determine the first and second level of digital divide in rural Bangladesh

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Abstract

This study aims to investigate the digital divide theory to provide a comparative dimension of internet disparities, and getting meaningful insights on the first (access to internet) and second (skills and utilization) level of digital divides in context of rural Bangladesh. The first-time representative survey of 6500 samples from rural Bangladesh explores the extent of differences in internet access, skills in navigating internet, and utilization of internet. The socio-demographic factors are identified to explain those inequalities using regression approaches. Findings reveal very clear-cut variations in outcomes. More specifically, age, gender, marital status, household size, educational qualification, household's income and geographical location variables are found to explain the disparity in internet access. Moreover, factors associated with the second level of digital divide measured in terms of web-browsing and information seeking skills also vary within and across the categories though in some way it differs from the first level of digital divide. The internet skills variable is found to be a statistically significant factor in explaining variations in internet utilization.

Keywords Digital divide · Access to internet · Internet skills · Internet usage · Rural Bangladesh

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Introduction

Disparities in individual's access to internet facilities is referred to the first level of digital divide whereas internet skills and usage gap among the people are embedded in the notion of the second level of digital divide (Dimaggio et al. 2004; Newhagen and Bucy 2003). Recently, the discussion surrounding the digital divide has focused on the internet skills, denoted as second level of digital divide, is crucial for effectively and efficiently utilizing internet. Therefore, inequality in internet access and internet skills are embedded in the digital divide jurisdiction, which depends on the idea that there are both positive and negative consequences of access and non-access to internet infrastructure respectively. The theory of digital divide is used in this study to understand different forms of internet inequalities and providing a meaningful insight into the spectrum of individuals' access to internet, internet skills and utilization in the rural context of Bangladesh due to the presence of dearth of facts and a research gap too. The first-time nationally representative survey implemented by BRAC Institute of Government and Development (BIGD), BRAC University, Bangladesh, collected data from rural people on access to digital devices, internet access, online navigating skill and utilization of internet to explore the extent of gaps as well as to identify the factors associated with these kinds of disparities for policy implications.

Bangladesh is recognized as a country with a strong divide between its rural and urban areas, primarily driven by differences in economic status such as wealth, income etc. (Ullah 2020). Also, the digital divide or technology gap between 'haves and have-nots' intensifies such inequalities and results in other social, gender, and spatial inequalities (Islam and Inan 2021). This gives importance to the even diffusion of Information and Communication Technologies (ICTs henceforth) throughout the country to drive the access to information and knowledge as well as internet skills enhancement as uneven distribution of ICTs between rural and urban results in an uneven impact on their progress towards development (Brixiova et al. 2009). Bangladesh government has been giving importance to internet access and internet skills after realizing the uneven distribution of economic benefits derived from the unequal distribution of ICTs between rural and urban (Aziz 2020). Therefore, the government is providing efforts to reduce the rural-urban divide to ensure an even development through eliminating the rural-urban divide in digital access, denoted as first and second level of digital divides, which has been regarded a significant challenge for both policymakers and academics (Billon et al. 2009).

Though Bangladesh started internet usage in the early 1990s, it did not see any success in its growth and penetration until 2008 (Mazumdar and Alharahsheh 2020). Growth and penetration of internet access and usage was one of the lowest in the world and reached some way to only 0.4% of the population in 2008 (The World Bank 2024). Moreover, mobile internet inception in 2005 also did not bring any considerable improvements in internet access and usage during that period. However, Bangladesh experienced substantial growth in internet subscriptions, with nearly 3.5% of the population gaining internet access in 2011. Such improvements in internet access were the resulting outcomes of the two main factors: the significant reduction in the costs of internet-enabled devices and internet service facilities. Since 2011, a consis-

tent and impressive growth in internet usage has been observed, especially in rural regions, due to pro-rural policies by government such as the establishment of Community Information Centres (CICs), ensuring internet facilities in Local Government Institutes (LGIs) and public services offices, and the integration of ICTs into education, healthcare services, and NGO efforts. For example, the government of Bangladesh has already set up 5275 digital centres and 4550 Union Digital Centres (UDCs) to link rural residents with government internet services (Islam 2018; Islam and Tsuji 2011). More specifically, Bangladesh has reached an important milestone in enhancing internet accessibility through the implementation of 3G and 4G technologies (i.e., the driving forces of digitization in rural Bangladesh), in 2013 and 2018 respectively (Rahman and Sultana 2022). However, second level of digital divide in internet skills and usage continue to increase as per scholars of the second level of digital divide.

Despite such improvements in internet penetration, it poses a serious concern to ensure equitable internet access for the rural people of Bangladesh. The previous studies did not show any evidence on digital divide issues in terms of the internet access and internet skills dimension in context of rural Bangladesh. This study also contributes to provide information using a large scale nationally rural representative sample. Therefore, these issues have now become a key concern and challenge for Bangladesh. This BIGD's study is imperative for the Bangladesh to comprehend the disparities in internet access, skills and usage in rural Bangladesh. This study contributes here using micro-level data to help undertake successful policies for reducing the gaps in multi-dimensional digital divides, ensuring e-inclusion, and finally, reducing the rural-urban digital divide.

This micro-level internet study also contributes to the internet literature by exploring the likelihood of different covariates of the first and second levels of digital divides in the context of rural Bangladesh, along with the impact of these factors on internet skills and usage. More specifically, many covariates such as age of the respondents, household size, income, educational qualification, occupation, marital status, sex, and geographical location are used to assess the likelihood of these covariates on internet access and internet skills, respectively, using Probit regression models. Besides, the Ordinary Least Square (OLS) method is employed to figure the factors which determine internet usage among the rural people.

Literature review

The Internet of Things (IoT) is a network of interconnected devices, where these devices are wirelessly linked through smart sensors, potentially representing a new evolution of information and communications technology (ICT) (Li et al. 2015). Internet of Things (IoT), comprised of microprocessor and sensors including sound, movement and temperature, are connected to the internet for delivering profound benefits to internet users (Atzori et al. 2010; Gubbi et al. 2013), who depend in part on the internet knowledge, internet access, internet skills and informed usage for realizing such benefits (Van Deursen and Mossberger 2018). The extent of disparities in internet access, internet skills, and usage highlighted the concept of digital inequality, which has grown significantly over the last few decades, and policymakers have been

working in this area to address and explain the various dimensions of digital inequalities (Beaunoyer et al. 2020; Van Deursen and Van Dijk 2015, 2019; Reisdorf and Groselj 2017; Blank and Groselj 2014; Zillien and Hargittai 2009; Ono and Zavodny 2007; Helsper 2012). It is also found that disparities in internet access worsen social, health and economic inequality (Van Deursen et al. 2021; Katz et al. 2017; Latulippe et al. 2017; Ma et al. 2020).

Nowadays the terminology 'e-inclusion' is highly discussed everywhere as many disadvantaged people are lagging in terms of internet infrastructure such as internet access, internet skills, and digital literacy. Such digital divide termed as complex and multidimensional issues persists for many reasons such as obtaining internet access and internet skills are linked to several socio-demographic and spatial characteristics including income, age, gender, educational qualification, race, gender, geographical location, digital attitude, knowledge and skills (Yoon et al. 2020; Viollaz and Winkler 2020; Rahiem 2020; Carr 2007). Similarly, cross-country analyses reveal that education and income are the dominant covariates for the first and second levels of digital divides (Rotondi et al. 2020; Hilbert 2010). Despite the role of factors such as economic and education, age of an individual is crucial to measure the digital divide (Neves et al. 2018; Walker et al. 2020). Although socio-demographic and economic predictors determine access to internet, internet skills, types of usage and frequency of usage play a very significant role in internet skills. For example, people who use the internet for many years and go online more frequently have more knowledge about internet activities than those who use the internet for a few years and go online less frequently (Hargittai and Hinnant 2008). Digital literacy, media literacy, and information literacy are three interrelated issues in today's information-driven society (Koltay 2011). Despite the rapid growth of information technology, users still need the skills to effectively use and contribute to it (Mutiar and MKep 2021). Digital citizens need to integrate knowledge and attitudes with digital skills to actively engage in social life (Melhuish et al. 2018), and media literacy strengthens digital citizenship practices (Park et al. 2021). Media health literacy is also related to well-being of a person, because various aspects of media health literacy also influence individuals' health and behaviours (Hung et al. 2021).

Significant differences in internet access and internet skills persist between rural and urban in the European Union (EU) countries despite no such differences are evident for infrastructural indicators of the internet (Várallyai et al. 2015). Although gender is often considered a factor in the digital divide, experimental research that accounts for income, education, and employment as confounding variables reveals that women are more likely than men to have internet access and skills (Rubin 2017). DiMaggio and Hargittai (2001) have identified five dimensions of digital inequality investigating the internet penetration data: technical means, the autonomy of use, skills, social support, and the purpose of use. The above discussion highlights that one of the primary challenges for policymakers is to ensure inclusive access to internet and essential internet skills, thus fostering the effective utilization of ICTs. The digital divide debate has changed from a simple focus on internet access, internet skills and usage, and more recently, to the positive outcomes of internet use, a shift defined as the third-level digital divide (Wei 2012; Scheerder et al. 2017). People with lower educational attainment, limited digital problem-solving skills, insufficient

information literacy, and lower frequency of internet use face significant barriers to receive basic service from internet, identified as third level digital divide (Anrijs et al. 2022).

According to Bangladesh Telecommunication Regulatory Commission (BTRC), active internet connections (i.e., 50% penetration) in Bangladesh comprise 90.5 million in 2018. However, the 'After Access' survey conducted by LIRNEasia, a think-tank based in the Asia Pacific region, revealed in 2018 that Bangladesh has the lowest internet usage among the surveyed Asian countries, with only 13% of the population using the internet. However, this figure increases to 33% among the population aged between 15–65 years. On the flip side, Bangladesh performed better regarding access to mobile-phone, with 74% of the population aged 15–65 years having access to mobile-phone. This figure represents one of the highest percentages of mobile phone usage among the surveyed Asian countries. This differences in mobile penetration and access to internet indicates that a substantial difference in access to internet persists in Bangladesh. Additionally, Bangladesh faced significant gender disparities in internet and computer usage, along with mobile accessibility, ranking among the most pronounced globally. For instance, the gender gap in internet access is considerably higher (i.e., 62%) in Bangladesh compared to mobile phone ownership, which stands at 34%. Last but not least, the clear-cut divide between rural-urban in Bangladesh implies that rural dwellers lag behind the urban dwellers by 38% (LIRNEasia 2019).

Unfortunately, despite the growing attention of policymakers and researchers on the first and second levels of digital divides, evidences remain still unrevealed in context of Bangladesh, particularly for rural Bangladesh. Specifically, essential characteristics of the internet users such as household income, household size, educational qualification, occupation, age, marital status, sex, and geographical location are neglected to a great extent to explain the multi-dimensional digital inequalities. The contribution of this study is to provide empirical evidence on the first and second level of digital divides regarding internet access, online navigating skills and utilization of internet, respectively. While Bangladesh has ensured a moderate internet accessibility, but the evidence on internet skills and usage disparities are very limited. Therefore, we utilized the resource and appropriation theory of Van Dijk (2005) which explores multi-dimensional digital divide in internet access, skills and usage. The existing literatures on digital divide do not provide any evidence on the internet skills and usage disparity dimensions, with the narrowing access gap in context of developing countries like Bangladesh. Despite a lower gap in internet access, the second levels of digital divide generate inequality in taking digital services, resulting in unequal distribution of the benefit derived from digital technologies and services (Litt 2013). The higher the digital skills and capacity in internet browsing and usage, the greater the potential benefits from technology. This study explores three phases of internet adoption internet access, online navigating skill and utilization of internet to identify how the socio-demographic determinants are linked to the first and second levels of digital divide in rural Bangladesh.

Research questions

a. What socio-demographic factors contribute to access to internet in rural Bangladesh?
b. What factors contribute to the emergence of the second level of digital divide in internet skills and usage?
c. Does internet skills matter for internet usage?

Conceptual framework

The study utilizes Van Dijk's (2005) conceptual framework to analyse and elucidate the digital disparities in rural Bangladesh. The causal model identifies how access to the internet differs and creates a first-level digital divide, and partial analysis of Cumulative and Recursive Model (CRM) recognizes the second-level digital divide (Fig. 1).

Disparities in personal and socio-demographic attributes such as household size, different levels of income, education, age (young vs. old), marital status, gender, occupational categories, region (rural vs. urban) and spatial (divisions in our case) characteristics determine digital divide measured in terms of the access to the ICTs (Eastin et al. 2015; Hargittai 2010). Since such inequalities rely on the features of digital technologies, uneven adoption of technology leads to uneven participation in the society and overall well-being. Finally, inequalities in societal participation and wellbeing reproduces the inequalities in personal and socio-demographic characteristics. Adopting this model, we analyse how heterogeneous characteristics of individuals and households determine internet access, which creates first level of digital divide in rural Bangladesh. This is also reasonable to assume that young individuals and those with higher level of education have greater access to the internet than the older individuals and those with lower levels of education, respectively (Hargittai 2010). Regional variations, household size, gender, marital status, households' income and spatial locations may also cause variations in internet access and thus, we have analysed how the above-mentioned factors determine the internet access (Blank and Groselj 2016).

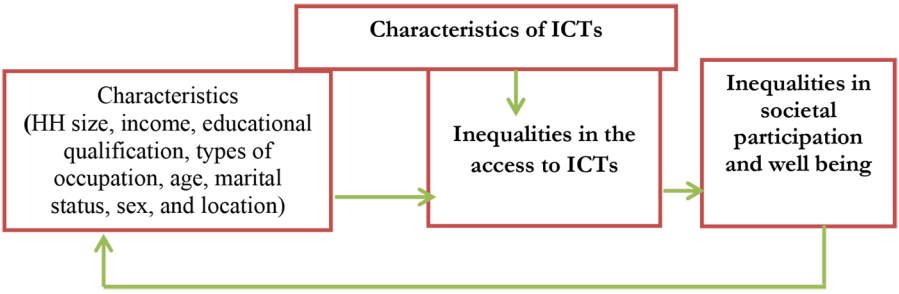


Fig. 1 A causal model. *Source* Adapted from van Dijk (2005)

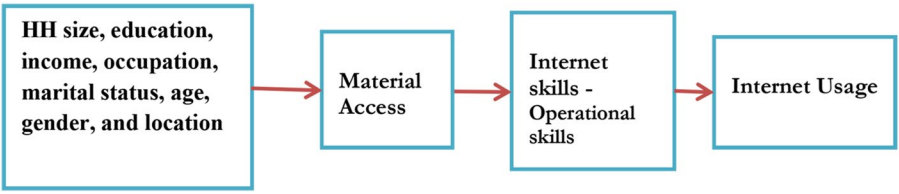


Fig. 2 A cumulative and recursive model. *Source* Adapted from van Dijk (2005, p. 22)

Van Dijk (2005) also formulated four consecutive phases regarding the accessibility of ICTs: motivational, material, skills and utilization (Fig. 2). For our analytical purpose, we have used the data on material access, skills and usage.

To explore the aspects of material accessibility, certain factors such as household size, level of income, educational qualification, types of occupation, location, age, gender, marital status, etc. are important to explain such access. After material access, skills of an individual to command and use of ICTs are important- operational skills, informational skills and digital skills (Hargittai 2010; Hargittai 2002). We tested only the operational skills under the survey. Personal and household's socio-demographic characteristics also matter for internet skills (Eynon and Helsper 2011; Friemel 2016). The final stage represents the internet usage determined by the internet skills. Individual's material access, operational skills have been analysed based on the findings.

In this research, internet accessibility is defined as possessing a broadband connection, utilizing mobile data from one's personal device or others' mobile phones, or accessing the internet from a nearby computer shop. For assessing operational and online navigation skill, participants were requested to download passport application document from the passport website of Bangladesh, locate information regarding passport application fees, and identify the hotline number for the passport office. Therefore, the examination of material access to the internet and internet skills in our results aids in identifying the factors contributing to variations. Lastly, the internet usage is modelled to assess the gaps based on the same covariates. Therefore, digital divide in rural Bangladesh is interpreted adopting the well-recognized framework of Hargittai (2001), has identified the significance of skills in explaining multi-dimensional digital divide.

Data and methods

Data

The study used nationally rural representative cross-sectional data, collected by BRAC Institute of Governance and Development (BIGD) from Bangladesh in 2019, to get a broader understanding on access to digital technologies and service taking behaviour. The 'Digital Literacy and Access to Public Services 2019' survey provides statistically representative data at two levels: (1) rural Bangladesh at the national level; (2) all administrative divisions of Bangladesh: Dhaka, Chattogram, Barisal, Khulna, Mymensingh, Rangpur, Rajshahi, and Sylhet. A total of 325 PSUs were ran-

domly selected using the probability proportion to size from each of the 8 strata, based on the Bangladesh population census of 2011. A robust and appropriate statistical approach were employed to randomly select 6500 households from 325 villages or primary sampling units (PSUs), with 20 households chosen from each PSU using the systematic random sampling technique. One of the main aims of this survey was to evaluate the status of internet accessibility, internet skills, and internet utilization among individuals in rural areas of Bangladesh (Shadat et al. 2020).

Explanatory variables

This study selected a better-informed member from a household and collected information about their age, household size, education, household's income, type of occupation, marital status, sex and location. Age variable is categorized into six: Below 25, 25–34, 35–44, 45–54, 55–64 & 65 and above. The household size variable has been also grouped into 5 categories. In addition, we recoded the independent variables in some places where appropriate.

Dependent or outcome variables

Three outcome variables were identified to explain the first and second levels of digital divide such as internet access, skills and usage. Regarding internet experiences of the rural dwellers in Bangladesh, the four key questions were asked to obtain the measure of internet access such as do you have a broadband connection at home? Do you have an internet connection on your mobile phone? Do you use or have access to other's mobile phone for accessing internet? Do you access internet from a nearby computer shop? we have generated 'internet access' dummy variable based on the responses to the four questions mentioned above (internet access = 1 or 0). Individual is considered to have internet access if they had access to any of the aforementioned internet facilities.

The study uses an established survey method to measure the internet skills (know-how to browse/use) using the online navigation test for knowing whether respondent was successful to download the passport document, application fee and passport hotline number through web-browsing. Based on the three questions whether the respondent could do the tasks, are used to generate a 'internet skills' dummy variable, with internet skills = 1 or internet skills = 0. We consider individuals to have internet skills if he/she could at least any of the three assigned tasks through internet browsing within 5 minutes.

For measuring internet usage, a list of activities such as reading news, searching information, downloading/listening songs, playing games, watching YouTube videos, paying bills, downloading/watching movies, online training, and use of social media were taken into account. Based on the internet activities performed by the individual, a composite variable representing the total number of activities was generated using the binary response variables for each activity (i.e., ranging from 0 to 9).

Methods of analysis

We explored the outcome variables include the internet access, skills and usage to analyse the first and second levels of digital divides in context of rural Bangladesh. The findings illustrate how each outcome variable correlates with the socio-demographic characteristics of the respondents, including age, household income, level of education, household size, occupation type, marital status, sex and geographical location. The regression models include the Probit model and Ordinary Least Square (OLS) regression model, which explain how the explanatory variables contribute to identifying disparities in the outcome variables.

The Probit econometric model is used to describe how internet access is determined. A random linear utility function is assumed to explain individuals' decision to have or have not internet access. The utility derived from internet access is assumed to be a linear function of individual's socio-economic and demographic variables (X_i) and unobservable and error measurements (ε_i). Hence the utility of an individual is related to household's having internet access ($U_{i,H}$) and not having it ($U_{i,N}$) can be described as:

$$U_{i,H} = X_i\beta_H + \varepsilon_{i,H}$$

$$U_{i,N} = X_i\beta_N + \varepsilon_{i,N}$$

An individual will choose to use internet if the utility derived from having internet access is greater than not having internet access. Let's define $Y_i = 1$ if the individual has internet access and $Y_i = 0$ if does not have. So, we can write $\Pr(Y_i = 1) = \Pr(U_{i,H} > U_{i,N}) = \Phi[X_i(\beta_H - \beta_N)]$ where Φ is define as the cumulative density function of error differences$(\varepsilon_{i,H} - \varepsilon_{i,N})$.

$$\Pr(Y_i = 1) = \Pr(U_{i,H} > 0) = \Phi[X_i\beta]$$

Similarly, we can also derive the probability function of internet skills. The model can be expressed as:

$$\Pr(Y_{i,s} = 1) = \Pr(U_{i,s} > 0) = \Phi[X_i\beta]$$

Under Probit regression model, the error terms are assumed normally distributed with mean 0 and variance 1 and it follows cumulative density function with the standard normal variable characteristic. So, our empirical approach to defining the regression model can be explicitly written as:

$$\Pr(Internet\ access) = \Phi[\alpha + \beta_0\, income + \beta_1\, education + \beta_2\, location + \beta_3\, gender + \cdots + \varepsilon_i]$$

Similarly, $\Pr(Internet\ skill) = \Phi[\alpha + \beta_0\, income + \beta_1\, education + \beta_2\, location + \beta_3\, gender + \cdots + \varepsilon_i]$

In the context of the OLS regression model, where the dependent variable (Y_i) captures the number of activities a person engages in online, the independent vari-

ables (X_i) represent the socio-demographic characteristics of that individual. The OLS model can be written as:

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_2 X_{3i} + \beta_3 X_{3i} \ldots \ldots \ldots + \beta_k X_{ki} + \varepsilon_i$$

Y_i: dependent variable (number of activities), $X_{2i}, X_{3i}, \ldots \ldots X_{ki}$: independent variable (socio-demographic characteristics), $\beta_2, \beta_3 \ldots \ldots \ldots \beta_k$: slope coefficients, measure the change in Y_i due to one unit change in X_i, and ε_i: error term.

Findings

Background characteristics

The following Table 1 provides us a snapshot on the socio-demographic, economic and spatial characteristics of the respondents. More than half of the respondents (55%) fall within the age range of 15–34 years. The participation of male (63%) as the respondent is higher than the female counterparts. Regarding educational qualifications, 18.11% have no formal education, while only 5% have attained a Graduate or equivalent level of education. The data also reflects the regional distribution of rural population by division, with a higher percentage of samples originating from Dhaka, Chittagong, and Rajshahi divisions, as per the 2011 census report. Approximately 39% of respondents report a family income below BDT 10,000, indicating a notably low economic status for these households. Furthermore, about 30% of the total sample is unemployed, with a majority being female respondents.

Internet access

An individual's internet access is defined as having access to at least one of the following methods of internet connection: broadband, mobile network, other's mobile phone or local computer shop. The results indicate that 37% of the participants have internet access. However, there is a notable disparity in the methods of internet access: approximately 33.6% of respondents access the internet through their personal mobile network, while the percentages for broadband, using others' mobile networks, and computer shops are 2.3, 2.5, and 0.3% respectively.

Internet skills

The practical measure of internet skills (know-how to search/browse) using practical assessment as described above, has been utilized to identify an individual's internet skills. We assess individuals' ability by applying the browsing test to download the passport application form, finding out passport application fee and passport office hotline number within the allocated five minutes. Skilled individuals are defined as those who were successfully completed at least one of the above-mentioned assigned tasks. When exploring internet skills, individuals who have access to the internet were taken into account. The results show a symmetric distribution of internet access

Table 1 Socio-demographic, economic and spatial characteristics of the sample respondents

Background characteristics	Percent
Age (in years)	
Below 25	32.61
25–34	25.95
35–44	19.94
45–54	11.11
55 and above	10.39
Household size	
1–2	12.36
3	20.63
4	27.67
5	19.21
6	10.64
>=7	9.49
Educational qualification	
No formal education	18.11
Below SSC	44.20
SSC/HSC/equivalent	32.43
Graduate or above	5.26
Monthly household income (BDT)	
Below 10,000	38.57
10,001–20,000	44.94
20,001–30,000	11.06
More than 30,000	5.43
Type of occupation	
Non-agriculture	33.74
Agriculture	17.82
Student	18.03
Unemployed/not in the labor force	30.42
Marital status	
Married	70.37
Not married	25.77
Others	3.86
Gender	
Male	63.09
Female	36.91
Location: division	
Dhaka	19.38
Chattogram	15.08
Rajshahi	15.08
Rangpur	14.15
Khulna	13.23
Barishal	8.00
Mymensingh	8.31
Sylhet	6.77
Total sample size	6500

among the respondents across the three tests used to measure internet skills. Among the respondents who have access to internet, only 35% of them have internet browsing skills. There is little variation observed across the three tests administered to individuals who have internet access, suggesting that those who were able to complete the first task were also capable of completing the subsequent tasks.

Internet usage

Internet access serve as the precondition for both internet skills and usage. However, only internet accessibility is insufficient for benefiting the users without active internet usage. To understand the usage level, respondents asked about their internet activities mentioned before. For understanding internet usage, individuals with internets access were included in our analysis. The results indicate that 31% of those with internet access don't engage in any of the nine internet activities. This implies that merely having internet access doesn't guarantee usage. About 15% of respondents have managed to perform four out of nine activities. When those who didn't perform any activities are excluded, the distribution shows a roughly normal shape, with the highest concentration around four activities.

Probit regression model: internet access and skills

Factors determine internet access

The findings from the Probit regression analysis (Table 2) for access to internet reveal how socio-economic and demographic factors determine their access to internet, defined as the first level of digital divide. Among these factors, age variable is found to be important predictor to explain the first level of digital divide in rural Bangladesh. Younger adults exhibit greater access to internet compared to their older counterparts, with age demonstrating statistical significance in predicting internet access. The marginal effect coefficient for age suggests that individuals under 34 are 18.5% points more inclined to have internet access compared to those aged 34 and older. Additionally, Household size variable exhibits significantly positive relationship with internet access, albeit with a negligible magnitude. Education variable found to a consistent predictor in explaining digital divide, reflecting the influence of both motivation and capacity to use internet, which align with the previous studies (DiMaggio et al. 2004; Van Dijk 2005), indicating that higher levels of education correlate with higher likelihood of access to internet (similar to Van Dijk 2005). In more detail, individuals with educational backgrounds spanning from Class I-VIII, SSC/HSC/equivalent, and beyond HSC levels show average probabilities of internet access approximately 13, 30, and 40% points greater, respectively, when compared to those without any formal education.

Economic status of the surveyed households, frequently measured by household income, is widely acknowledged as a significant predictor of individuals' access to internet (Katz and Rice 2002; Van Dijk 2005; Livingstone and Helsper 2007; Ono and Zavodny 2007; Goldfarb and Prince 2008). Probit results indicate that the likelihood of internet access increases as household income increases and thus, income

Table 2 Probit and OLS regression analyses on internet access, internet skills and usage

Variables	(1) Internet access	(2) Average marginal effect	(3) Internet skills	(4) Average marginal effect	(5) Internet usage
Age category (Coded 1 if age<=34 and 0, otherwise)	0.730***	0.185***	0.240***	0.080***	0.502***
	(0.048)	(0.012)	(0.0891)	(0.030)	(0.130)
Household size	0.024**	0.006**	0.00504	0.002	0.026
	(0.012)	(0.003)	(0.0164)	(0.005)	(0.025)
Educational qualification (Reference category: without formal education)					
Below SSC/equivalent	0.554***	0.132***	0.120	0.033	0.044
	(0.069)	(0.015)	(0.185)	(0.050)	(0. 244)
SSC/HSC/Equivalent	1.131***	0.302***	0.620***	0.200***	0.834***
	(0.072)	(0.017)	(0.184)	(0.050)	(0.246)
Above HSC	1.449***	0.399***	1.049***	0.362***	1.684***
	(0.102)	(0.028)	(0.198)	(0.057)	(0.274)
Income group (Reference category: BDT below 10,000)					
10,001–20,000	0.327***	0.085***	0.0881	0.029	0.116
	(0.044)	(0.011)	(0.0721)	(0.024)	(0.111)
20,001 to 30,000	0.774***	0.211***	–0.00888	–0.003	0.493***
	(0.066)	(0.018)	(0.0909)	(0.030)	(0.139)
>30,000	1.004***	0.276***	0.222**	0.075**	0.793***
	(0.090)	(0.025)	(0.108)	(0.037)	(0.168)
Occupation type (Reference category: Agriculture)					
Student	0.594***	0.149***	0.289**	0.097**	0.864***
	(0.088)	(0.023)	(0.127)	(0.041)	(0.193)
Unemployed	0.661***	0.167***	0.158	0.052	0.342*
	(0.080)	(0.020)	(0.132)	(0.042)	(0.198)
Non-agriculture	0.376***	0.091***	0.104	0.034	0.243
	(0.060)	(0.014)	(0.114)	(0.036)	(0.168)
Unmarried (=1 if unmarried and 0, otherwise)	0.426***	0.108***	0.103	0.034	0.391***
	(0.065)	(0.016)	(0.0831)	(0.029)	(0.129)
Gender (=1 if male and 0, otherwise)	0.698***	0.177***	0.265***	0.089***	0.920***
	(0.061)	(0.015)	(0.0815)	(0.027)	(0.128)
Division (Reference category: Rangpur)					
Barisal	0.578***	0.136***	0.429***	0.153***	0.086
	(0.089)	(0.021)	(0.141)	(0.050)	(0.227)
Chattogram	1.063***	0.267***	–0.118	–0.039	0.313*
	(0.076)	(0.018)	(0.117)	(0.039)	(0.185)
Dhaka	0.816***	0.199***	0.148	0.051	0.134
	(0.072)	(0.017)	(0.114)	(0.039)	(0.182)
Khulna	0.682***	0.163***	0.117	0.041	0.452**
	(0.076)	(0.018)	(0.121)	(0.041)	(0.193)
Mymensingh	0.508***	0.118***	0.345**	0.123**	−0.303
	(0.089)	(0.021)	(0.146)	(0.052)	(0.233)

Table 2 (continued)

Variables	(1) Internet access	(2) Average marginal effect	(3) Internet skills	(4) Average marginal effect	(5) Internet usage
Rajshahi	0.591***	0.140***	–0.131	–0.043	0.331
	(0.075)	(0.017)	(0.121)	(0.040)	(0.191)
Sylhet	0.474***	0.110***	–0.663***	–0.188***	–0.559**
	(0.095)	(0.023)	(0.177)	(0.047)	(0.245)
Internet skills (Coded 1 if individual with Internet skills and 0, otherwise)	No		No		1.421***
					(0.093)
Constant	–3.593***		–1.598***		–0.343
	(0.125)		(0.247)		(0.353)
Pseudo R^2	0.317		0.0931		
R^2					0.305
Observations	6500		2405		2405

Note dy/dx for factor levels is the discrete change from the base level; standard errors in parentheses; *** $p<0.01$, ** $p<0.05$, * $p<0.1$

creates the first-level digital divide. The probability of having internet access increases by 8.5, 21.1, and 27.6% points for individuals in the income groups of BDT 10,001–20,000, BDT 20,001–30,000, and more than BDT 30,000, respectively, compared to those in the income group below BDT 10,000. Occupation type also play role in determining access to the internet of an individual. Comparing individuals' involvement in agriculture as a reference group, those categorized as students, non-agricultural workers and unemployed or not in labour force individuals have greater access to internet. While there has been a decline in the gender gap in internet access in numerous developed nations, inequalities remain prevalent in developing countries such as Bangladesh. The 'gender' variable is also found to be statistically significant, indicating that males are approximately 18% points more internet access compared to females, proving the existence of a significant digital divide based on gender. Similarly, unmarried individuals show higher possibility (10.8% point) of accessing internet compared to the other individuals. Moreover, location of the individual is considered to be a significant variable in explaining internet access in rural Bangladesh. While some literatures define a divisional location in rural and urban terms, we investigated the digital disparities within the rural areas across eight divisions in Bangladesh to assess the degree of inter-divisional disparities in internet access. Using Rangpur as the reference division (which is identified as most poverty-prone division), it become apparent that significant divisional disparity in internet access persist. Compared to the reference division, the analysis reveals that Chattogram division exhibits the highest likelihood of internet access, with a 26.7% point increase. This is followed by Dhaka (19.9% points), Khulna (16.7% points), and Rajshahi (14% points), respectively.

Determinants of internet skills

Probit regression results of internet skills (Columns 3 and 4 of Table 2) explain how socio-economic characteristics are associated with individuals' internet skills and quantify the extent of disparities in navigating internet (the second-level of digital divide) in Bangladesh. The age of individual is also found to be a significant determinant of the second level of digital divide, significantly impacting the internet skills. Young individuals, aged below 34, are 8% points more likely to become skilled in internet usage compared to the older age group. However, the household size and marital status variable do not exhibit any influence on the acquisition of internet skills. We found no significant difference in internet skills between individuals with no education and those with education below the SSC level. However, the positive and significant correlation between internet skills and educational qualification of SSC and above suggests that a minimum level of education is required to have internet skills. While no significant association is found between monthly incomes up to BDT 30,000 and internet skills, incomes exceeding this threshold demonstrate a positive correlation with internet skills. Thus, the likelihood of having internet skills increases once an income threshold is reached.

Individuals who are students, non-agricultural workers, or unemployed have significantly better internet access than those working in agriculture, the marginal effect of student category was found to be statistically significant for internet skills variable also. The gender variable significantly determines internet skills in context of rural Bangladesh. Male individual is 8.9% point more likely to have practical internet skills compared to the female. Results for location are not the same as access to the internet. Individuals from Barisal and Mymensingh division exhibit better internet skills, with 15.3 and 12.3 higher percentage point, respectively, compared to those from reference division (Rangpur). In contrast, Sylhet division significantly lags behind other divisions. Hence, there is a regional perspective to existence of the second level of digital divide.

OLS regression results: internet usage

(Column 5 in Table 2) shows how internet usage differs across observable characteristics. More specifically, OLS findings complement binary analyses by illustrating which covariates are significant in explaining the disparities in utilization of internet while other covariates considered to be constant. The regression model confirms that internet competency is very important determinant for internet usage. Specifically, individuals with internet competency engage in 1.42 more online activities on average compared to those without this competency. Age variable is found to be significant, indicating that individuals in the younger age bracket (<=34 years) utilize the internet more than their counterparts, aligns with the previous studies (Barrantes and Vargas 2019). Therefore, internet usage gap by age persists in rural Bangladesh. However, household size remains insignificant. The findings of education variable align with those obtained for internet skills, indicating that individuals with SSC/HSC/equivalent and above HSC levels utilize internet more compared to the reference group, with the coefficient 0.83 and 1.63, respectively. Additionally, household

income levels significantly influence internet usage in rural Bangladesh. Individuals belong to the higher income group (specifically more than BDT 20,000) emerges as the important factor in explaining the extent of internet usage. In case of occupation type, the number internet activities are higher among the students and unemployed by 0.86 and 0.36 number, respectively, compared to those engaged in agricultural sector. Additionally, this study reaffirms the existence of a gender gap in internet usage, with males significantly utilizing internet more than the females. This study utilizes marital status as a determinant of internet usage and reveals that unmarried individuals engage in more internet activities compared to the married individuals. Regional differences in internet usage are not notable across several divisions, except for Khulna and Sylhet. Individuals from Khulna division notably engages in more internet activities compared to those Rangpur division. In contrast, individuals from Sylhet division perform less internet activities by 0.55 number in comparison to those from Rangpur, one potential explanation is differences in infrastructure and connectivity. Khulna may have better-developed internet infrastructure and connectivity, leading to higher internet usage rate by 0.45 number.

We performed the Breusch-Pagan (BPG) test to examine the heteroskedasticity in OLS regression model, this test fails to reject the null-hypothesis of homoscedasticity at 10% level of significance. So, the OLS regression model does not exhibit heteroskedasticity in the residuals. Additionally, the variance inflationary factor (VIF) is conducted to examine multicollinearity in the regression model among the explanatory variables. The average VIF is calculated to be 2.96, significantly below the threshold value of 10, indicating the absence of multicollinearity issue.

Discussion and conclusion

The study has two goals which are addressing the first and second level of digital divide concerning internet accessibility, skills and usages among the rural residents in the rural Bangladesh. In the developed countries, 87% of people have internet access, compared to 47% in developing nations, and only 19% in the least developed countries. So, lower nternet access in the developing and least developed countries suggests a higher likelihood of internet disparity within society (ITU 2019). Firstly, this paper investigates whether and how internet access varies across observable characteristics of the respondents. The findings show a clear-cut first level of digital divide in internet accessibility among the various groups examined in the study (Eynon and Helsper 2011; Friemel 2016). More specifically, younger people, those belong larger household, individuals with higher levels of education, student, those with higher incomes, unmarried individuals, males and those located in Chattogram and Dhaka exhibits greater level of internet access. Education variable found to a consistent predictor in explaining digital divide, reflecting the influence of both motivation and capacity to use internet, which align with the previous studies (Katz and Rice 2002; DiMaggio et al. 2004; Van Dijk 2005), indicating that higher levels of education correlate with higher likelihood of access to internet (similar to Van Dijk 2005). The positive and significant correlation between internet skills and usage and educational qualification of SSC and above suggests that a minimum level of education

is required to have internet skills. The gender variable was found to be a significant predictor of internet access, skills, and usage, implying a multidimensional gender digital divide (Antonio and Tuffley 2014). Comparing developed, developing, and least developed countries (LDCs), the gender disparity is highest in LDCs (42.8%), followed by developing countries (22.8%), and lowest in developed countries (2.3%) (ITU 2019).

Secondly, this study assesses whether rural individual with internet access exhibit disparities in the second level of digital divide measured by 'know-how to browse/search' (i.e., internet skills and usages). The findings reveal variations in both internet skills and usage within and across the analysed categories. The results provide us clear evidence that the internet access does not ensure equal skill and usage capacity (Blank and Groselj 2016). Internet skills is necessary for usage and getting benefit from technology (Ragnedda and Kreitem 2018). So, this study supports the theory of digital inequality in terms of internet access, internet skills and internet usage, illustrating how they differ across socio-demographic and economic characteristics (Islam and Tsuji 2011). The disparity in digital skills between developed and developing countries is 19% points for basic skills and 29% points for standard skills, with the gap being notably wider in the poorer countries (ITU 2018).

The study findings suggest that only material access to internet may not be sufficient to address digital divide in Bangladesh. For this purpose, relevant policies for different age, education, occupation, income and gender groups along with spatial locations are necessary to reduce the inequalities in internet access, skills and usage.

Limitations of the study

The findings of the research paper are based the cross-sectional data, opens up the window for further research. The panel data analysis would fit best to understand the dynamics of the determinants over time. Adding the digital skills competence domains such as problem-solving capacity, content creation and eGovernment service in assessing internet skills would provide us a deeper understanding of digital inequality. The quality of device, specification, internet quality, which are not taken into account in the analysis, also matter in the long-run for internet skills and usage.

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Data availability Data will be available upon request.

Declarations

Ethical approval Ethical approval was not be necessary because we used secondary data that is anonymized and cannot be traced back to individuals. The use of such data does not pose any risk to the privacy or well-being of individuals.

Informed consent Permission was sought from respondents as to whether they would like to participate in the study, and their participation was taken into consideration once they gave their consent.

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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